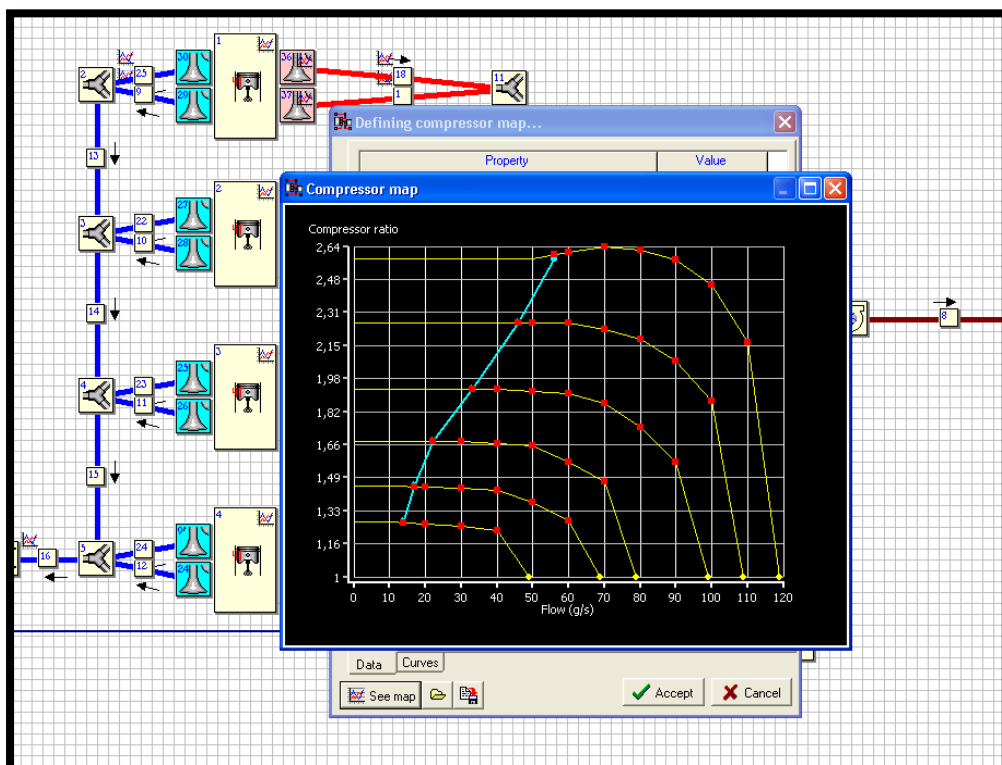


CRITT M2A

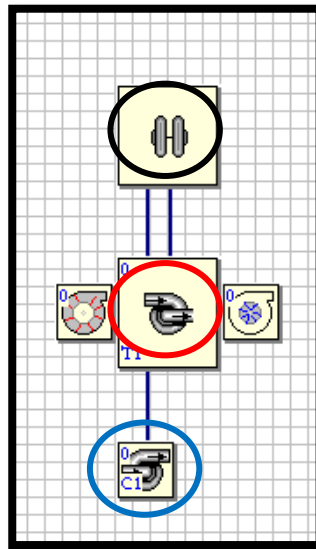
Tutorial: Modeling a Turbocharger in OpenWAM 2.0.2



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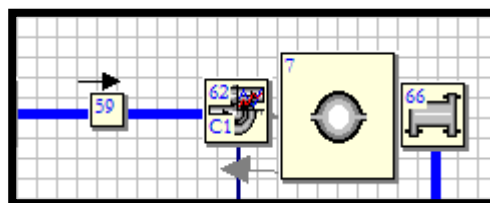
Setting up a turbocharger:

In the OpenWAM workspace, go to Auxiliar. Choose the first element, the turbocharger. A dialog box will open asking if you want OpenWAM to automatically generate a compressor and a turbine. Click Yes.



Your turbocharger has been created. The black element represents the turbocharger, the blue element the compressor, and the red element the turbine.

To connect the compressor to a duct, we recommend the following architecture:



Its outlet must be connected to a small volume (plenum). The compressor model used in this case is much more stable.

For the turbine, there are no special requirements.

The turbocharger element lets you define its rotation speed and its inertia. It doesn't connect to anything else. To model a compressor on its own, the software needs a complete turbocharger element; the turbine will only be connected, not defined.

Building a compressor map:

This method, which is fairly tedious, lets you fill in the compressor map in OpenWAM. Adding an Excel macro would save significant time and improve precision.

Property	Value
Map type	WAMer
Reference map pressure (bar)	0
Reference map temperature (°C)	0
Minimum speed (rpm)	0
Maximum speed (rpm)	0
Speed shift between curves (rpm)	0
Minimum corrected mass flow (g/s)	0
Maximum corrected mass flow (g/s)	0
Curves mass flow interval (g/s)	0

Note: Reference map pressure (bar)

Data Curves

See map Accept Cancel

Here is the dialog box you'll see when building a compressor map.

- Map Type: two types are available, WAMer and SAE; here we'll show you the WAMer map. The WAMer map lets you enter each πC and efficiency value for each iso-speed line. The SAE map must be entered as a table.

- Reference map pressure: enter the reference pressure used when building the map.
- Reference map temperature: enter the reference temperature used when building the map.
- Minimum speed: enter the lowest iso-speed in your map.
- Maximum speed: enter the highest iso-speed in your map.
- speed shift between curves: enter the iso-speed step used to build your map. Example: a compressor map from 100,000 rpm to 225,000 rpm with 25,000 rpm steps.
- Minimum corrected mass flow: enter the lowest mass flow in your map. Note: we recommend using the default value of 0, otherwise the model may fail to compile.
- Maximum corrected mass flow: enter the highest mass flow in your map.
- Curves mass flow interval: enter the mass flow step you want, based on the precision you're after. Example: for a mass flow from 0 to 100 g/s with 10 g/s steps, that gives a maximum of 10 points per iso-speed. A 5 g/s step gives a maximum of 20 points per iso-speed.

100000rpm

Property	Value
Surge point mass flow (g/s)	0
Surge point compression ratio	0

Property	Value
Mass flow for Rc=1 (g/s)	0

Mass flow (g/s)	Compression ratio
0	0
10	0
20	0
30	0
40	0

Mass flow (g/s)	Efficiency
0.	0.

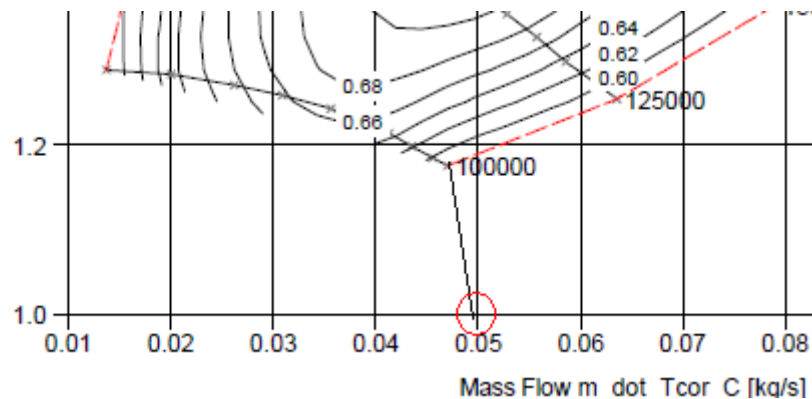
Accept Cancel

In the "Curves" tab, you can start filling in your flow/compression-ratio points. We recommend having a paper copy of your compressor map on hand.

- Surge point mass flow: enter the mass flow at the surge point for this iso-speed
- Surge point compression ratio: enter the compression ratio at the surge point for this iso-speed.

- Mass flow for $R_c=1$: Enter the extrapolated value for when your iso-speed curve crosses the compression-ratio axis at 1.

The $R_c=1$ point must take into account the calculation step provided. So, for example, if your point is at 60 g/s, you'll need to enter the intermediate flow/compression-ratio points that aren't plotted on the curve, as shown in the figure below.

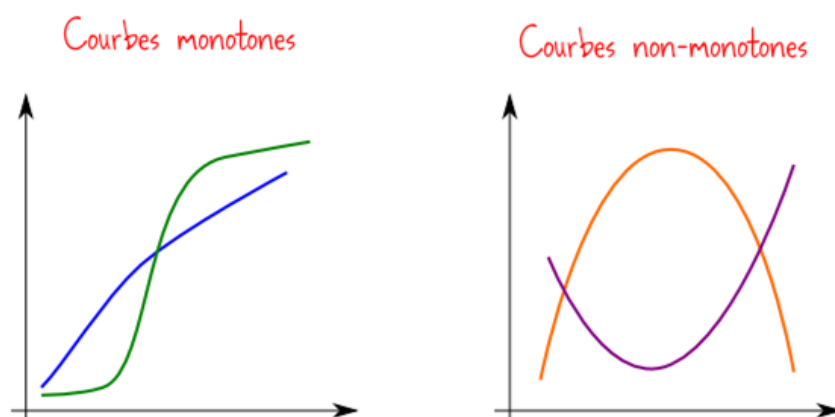


- Compression ratio: Enter the compression-ratio values corresponding to the mass-flow values provided by the software.

- Efficiency: Finally, you'll need to fill in the compressor efficiency table. To do this, click the black cross. It lets you add rows to your table. To fill in the mass-flow values, we recommend using the values the software provided in the previous table. Also add the surge point.

Warning

During simulation, problems can arise with the compressor map. To avoid this, make sure the compression-ratio values at two consecutive points are never equal, and the same for the efficiency values. Also, your map must trace a strictly monotonic curve. Otherwise, this can cause problems when simulating the model.



If your flow/compression-ratio table includes the surge point and the $R_c=1$ point, the map can cause problems. We recommend defining them so that the table excludes these two values.

Building the turbine map:

For the turbine map, the process is a bit less complicated.

Right-click the turbine, go to turbine then data. The following dialog box opens:

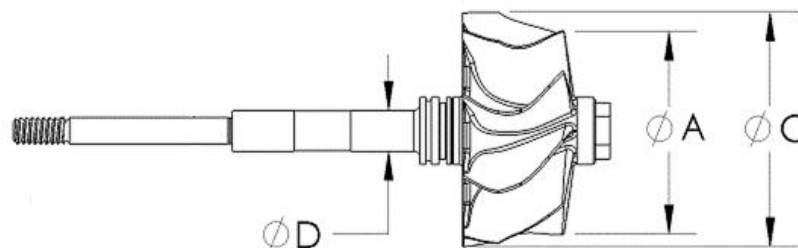
Property	Value
Turbine type	Map
Wheel diameter (mm)	36
Obtaining efficiency	Real curve (3rd degree polynomial)
Wheel output diameter (mm)	30
Nut diameter (mm)	9
Turbine inlet diameter (mm)	24
Critical angle (°)	0,068
Number of points	1
Initial rack position	0

Note: Turbine diameter (mm)

Data Map Control

Accept Cancel

Turbine type: you can choose between Map, fixed geometry, and variable geometry. Selecting fixed geometry ignores the turbine map you entered. To use an existing map, you must therefore select Map as the turbine type.



Wheel diameter: wheel diameter. $\varnothing C$

Wheel Output diameter: turbine outlet diameter $\varnothing A$

Nut Diameter: nut diameter. $\varnothing D$

Turbine inlet diameter: turbine inlet diameter

Critical angle: this is a parameter used for VGTs (variable-geometry turbines), corresponding to the stator blade angle at which the stator vanes are tangent to the rotor. If you're using a fixed-geometry turbine (FGT), you can enter 90°.

Number of points: for a variable-geometry turbine, select the number of vane positions based on how many positions it has.

Initial rack position: for a variable-geometry turbine, the starting map among your number of points.

In the Map tab, you can enter your turbine map. First, you'll see a drop-down list letting you enter a different map depending on the rack position.

For a fixed-geometry turbine, skip this step. For a variable-geometry turbine, you'll need to give the angle based on the vane position.

The map to enter is of the SAE type. To make this easier, you can use a text editor (Notepad). Here is an example of a (partial) turbine map. The first number represents the "rack position". The number "60" represents the number of data rows. "1" represents the rack position, and "2.500000e+01" the vane angle for this map. Then comes the data.

You'll need to fill in this data at the top of the file to get the data to compile correctly.

Warning

Don't forget to use a period as the decimal separator. Each column must be separated by a space. If a turbine error is returned during compilation, check that the map is strictly monotonic with respect to the expansion ratio. Check the other data if you're still having problems. If your text file doesn't follow this exact format, OpenWAM will have trouble reading it, which will result in a compilation error.

Here is the format of the final text file. The first column represents the turbocharger speed, the second the expansion ratio, the third the corrected mass flow, and the last the efficiency.

```
1
60 1 2.500000e+01
5.64E+01 1.26E+00 4.64E+00 4.55E-01
5.64E+01 1.29E+00 4.85E+00 5.06E-01
5.64E+01 1.31E+00 5.04E+00 5.25E-01
5.64E+01 1.33E+00 5.19E+00 5.34E-01
5.64E+01 1.34E+00 5.28E+00 5.42E-01
5.64E+01 1.35E+00 5.31E+00 5.47E-01
5.64E+01 1.35E+00 5.34E+00 5.50E-01
5.64E+01 1.36E+00 5.35E+00 5.54E-01
5.64E+01 1.36E+00 5.37E+00 5.54E-01
5.64E+01 1.36E+00 5.39E+00 5.54E-01
7.06E+01 1.39E+00 5.31E+00 5.34E-01
7.06E+01 1.44E+00 5.54E+00 5.80E-01
```